

Timing Equity Quant Positions with Short-Horizon Alphas

VINESH JHA

VINESH JHA
is CEO at ExtractAlpha
in Hong Kong.
vinesh@extractalpha.com

In this article, we demonstrate some simple strategies for using relatively fast-moving alphas to improve the timing decisions of systematic equity portfolios based on slower-moving alpha signals.

Many managers are aware of the alpha in short-horizon signals but do not use these alphas due to their high turnover; the portfolio may be large, or the mandate may not call for mid-frequency trading. We find that short-term alphas can be used by such managers to time trades that a longer-horizon strategy would enter regardless, thereby improving the price of the entry and exit points without incurring incremental transaction costs.

We demonstrate that a basic fundamental and momentum strategy with annual net returns of 5.5%, a net Sharpe ratio of 0.55, and daily turnover of 6.4% can be improved to annual net returns of 10.0% and a net Sharpe ratio of 0.97, with a slight *reduction* of turnover, by implementing simple entry and exit rules based on a short-horizon alpha. The value added is very consistent over time, offers drawdown protection in volatile markets, and survives reasonable transaction cost and latency assumptions.

Short-horizon alphas can thus be effective as *tactical* overlays that can improve risk-adjusted returns without unduly influencing the underlying *strategy*. The implementation could be as straightforward as a daily

pre-trade screen on position entries and exits prior to the open.

THE FACTORS

We begin with a simple example strategy: a long–short decile strategy built with an equally weighted factor blend of basic value, profitability, and momentum alphas, rebalanced daily, with a daily turnover of 6.4%.

- *Value* is defined as an equal blend of industry-relative trailing earnings yield and trailing sales yield.
- *Profitability* is defined as the industry-relative return on total assets (ROA).
- *Momentum* is defined as the industry-relative total return of the stock from 12 months ago to 1 month ago.

Our base strategy is not meant to be comprehensive or even particularly powerful—in fact over our time period, it has very modest returns net of costs—as we are mostly concerned with whether it can be improved through the use of timing.

We then examine the use of short-term alphas for timing the trades of this base strategy. In particular, we examine the Tactical Model (TM1), a cross-sectional stock selection score that combines several short-horizon factors (Jha [2016a]).

- *Reversal*. Single-stock returns tend to reverse over a one-month horizon (De Bondt and Thaler [1985]). The reversal effect is stronger when measured using returns residualized to common risk factors (Blitz et al. [2013]); and it is weaker when information-driven than when liquidity demand-driven (Avellaneda and Lee [2010]).
- *Factor momentum*. Stock idiosyncratic returns tend to reverse, but their common factor components tend to trend (Hameed [1997]).
- *Liquidity shock*. Stocks that have short-term increases in liquidity command additional attention from investors and become more accessible to investors with short-sale constraints (Rohal et al. [2015]; Bali et al. [2014]).
- *Seasonality*. Stocks that have historically outperformed during a particular time of year in past years tend to continue to outperform at that same time of year in subsequent years (Heston and Sadka [2008]).

All results in this article are market neutral, with portfolios that are long the top decile and short the bottom decile of a particular alpha rank, with the returns being a percentage of one side; that is, the market-neutral return is the difference between the long return of the long portfolio and the long return of the short portfolio. All portfolios are sourced from a liquid universe of U.S. equities, updated daily, with market caps over USD 100 million, USD 1 million average daily trading volume, and \$4 nominal price. Both our base strategy and TM1 are formulated as integer percentile rankings within this universe, with 1 representing the strongest short signal and 100 representing the strongest long signal. Our tests run from January 2000 through December 2015. One-way transaction costs are assumed to be 2 basis points (bps) for large caps (the largest 500 stocks by market cap), 5 bps for mid caps (the next 500), and 10 bps for small caps (the rest). The source price, volume, fundamental, and industry data for all factors were provided by FactSet.

SIMPLE BLENDS

There is relatively little academic literature that gives practical advice for managing a dynamic portfolio in the presence of both long- and short-term forecasts. Qian et al. [2007] advocated blending short- and

EXHIBIT 1

Portfolio with 70% Base Strategy/30% TM1

	Annualized Gross		Annualized Net		Daily Turnover
	Return	Sharpe	Return	Sharpe	
Base	10.8%	1.08	5.5%	0.55	6.4%
Blend	40.3%	4.01	17.2%	1.70	28.8%

Source: *ExtractAlpha*.

long-term predictors and also using lagged short-term predictors to increase their effective horizon. This approach has the undesirable characteristics of both increasing the turnover of the long-horizon strategy and not fully capturing the efficacy of the short-horizon strategy. Exhibit 1 shows what happens when we run a portfolio based on a simple blend consisting of 70% base strategy and 30% TM1.

We can see that the blend is, in fact, more powerful, but it fundamentally changes the characteristics of the long-horizon portfolio. In addition to the increased turnover, the risk profile of the portfolio also changes. For example the blend alpha's top decile has an exposure to the value risk factor of only 0.4, versus 0.7 for the base alpha (in *z*-score terms). Therefore a *strategy* blend does not seem to be the way to go. Next, we examine a more tactical approach.

TACTICS

We apply the following two methodologies, each of which could simply be applied in the form of a daily screening process prior to the trading day.

1. *Entry management*: If the base strategy would ordinarily enter a long (short) position and TM1 has a significantly low (high) score, *delay the entry* until the TM1 score is no longer very low (high).
2. *Exit management*: In addition, if the base strategy would ordinarily exit a long (short) position and TM1 has a significantly high (low) score, *delay the exit* until the TM1 score is no longer very high (low), but for no longer than 30 calendar days.

We could employ a mild version of entry and exit management where we allow TM1 to merely *veto* the trades the base strategy would otherwise do; or, we

could be much more aggressive and require that TM1 in fact *corroborate* or confirm those trades. A corroboration approach is more like performing a screen on both the base strategy and TM1. Here, we look at a spectrum of TM1 thresholds that span from veto through corroboration.

Exhibit 2 shows gross and net returns and Sharpe ratios for our base strategy and then by adding entry and exit rules for varying levels of TM1, from veto (long 5/short 96) to corroborate (long 90/short 11).

TM1 significantly improves returns and risk-adjusted returns for all implementations, with no significant increase in turnover or decrease in number of positions. The more aggressive we are about ensuring that

TM1 lines up with the base strategy direction, the more powerful the results. The turnover after implementing both entry and exit rules is actually somewhat lower than the baseline turnover of 6.4%, as on average, we delay exits from those positions we do end up entering.

In Exhibit 3, we plot the cumulative gross and net returns for the base strategy and varying levels of entry and exit rules based on TM1.

Exhibit 4 shows the value added in terms of increased net returns and Sharpe ratios and decreased turnover for the 20/81 implementation of TM1 entry and exit points.

Value is added across all years except 2011, and across capitalization ranges for the three metrics (with the

EXHIBIT 2

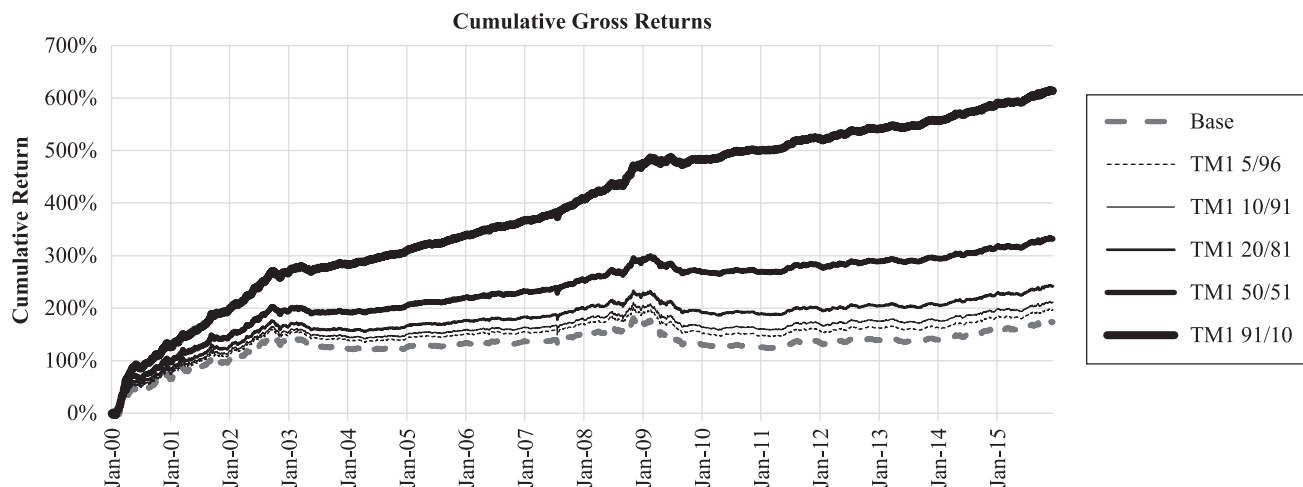
Returns and Sharpe Ratios for Base Strategy and for Varying Exit and Entry Levels of TM1

	Annualized Gross		Annualized Net		Daily Turnover	Stocks per Side
	Return	Sharpe	Return	Sharpe		
Base	10.8%	1.08	5.5%	0.55	6.4%	252
Long 5, Short 96	12.3%	1.19	7.0%	0.68	6.4%	227
Long 10, Short 91	13.2%	1.28	7.9%	0.77	6.3%	227
Long 20, Short 81	15.1%	1.47	10.0%	0.97	6.1%	228
Long 50, Short 51	20.8%	2.05	16.2%	1.60	5.6%	232
Long 90, Short 11	38.5%	3.71	33.8%	3.26	5.6%	220

Source: ExtractAlpha.

EXHIBIT 3

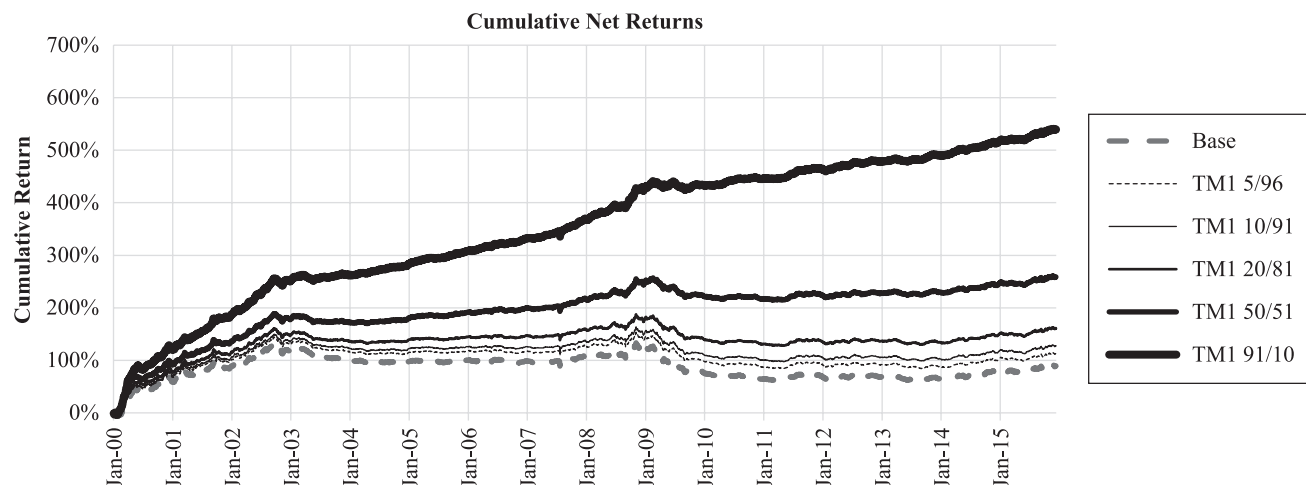
Cumulative Returns for Base Strategy and for Varying Exit and Entry Levels of TM1



(continued)

EXHIBIT 3 (continued)

Cumulative Returns for Base Strategy and for Varying Exit and Entry Levels of TM1



Source: ExtractAlpha.

EXHIBIT 4

Value Added for the 20/81 Entry and Exit Points for TM1

	Base Strategy			TM1 Enter/Exit at 20/81			Value Added		
	Net Ret	Net Sharpe	Turnover	Net Ret	Net Sharpe	Turnover	Net Ret	Net Sharpe	Turnover
Overall	5.5%	0.53	6.4%	10.0%	0.97	6.1%	4.5%	0.44	0.3%
2000	71.0%	5.09	7.5%	90.2%	6.47	7.5%	19.2%	1.38	0.0%
2001	17.2%	1.12	7.9%	25.6%	1.72	7.9%	8.4%	0.60	0.0%
2002	34.8%	2.51	7.0%	38.7%	2.83	6.7%	3.9%	0.32	0.3%
2003	-17.3%	(2.03)	6.8%	-11.9%	(1.40)	6.6%	5.4%	0.63	0.2%
2004	-7.7%	(1.21)	5.7%	-1.9%	(0.31)	5.6%	5.8%	0.90	0.1%
2005	4.3%	0.75	5.9%	7.4%	1.30	5.6%	3.1%	0.55	0.3%
2006	-1.4%	(0.21)	5.5%	2.2%	0.34	5.2%	3.6%	0.55	0.3%
2007	9.9%	0.96	6.7%	13.5%	1.28	5.8%	3.6%	0.32	0.9%
2008	15.3%	0.95	6.8%	20.3%	1.32	6.4%	5.0%	0.37	0.4%
2009	-43.5%	(2.79)	7.3%	-36.5%	(2.40)	7.1%	7.0%	0.39	0.2%
2010	-14.3%	(2.80)	6.5%	-11.4%	(2.20)	6.3%	2.9%	0.60	0.2%
2011	6.5%	0.99	6.1%	6.3%	0.96	6.1%	-0.2%	(0.03)	0.0%
2012	1.9%	(0.27)	5.6%	-0.4%	(0.05)	5.5%	1.5%	0.22	0.1%
2013	-3.5%	(0.63)	5.5%	-2.3%	(0.42)	5.2%	1.2%	0.21	0.3%
2014	10.6%	1.40	5.5%	11.3%	1.51	5.2%	0.7%	0.11	0.3%
2015	13.0%	1.40	5.9%	14.4%	1.56	5.3%	1.4%	0.16	0.6%
Large Cap	3.4%	0.21	9.3%	8.4%	0.53	10.4%	5.0%	0.32	-1.1%
Mid Cap	16.4%	1.01	9.7%	20.2%	1.24	9.5%	3.8%	0.23	0.2%
Small Cap	2.4%	0.21	7.8%	7.1%	0.61	6.4%	4.7%	0.40	1.4%

Source: ExtractAlpha.

exception of large-cap turnover), including 2009 when our sample strategy experienced a large drawdown due to its momentum strategy; TM1's typically strong performance in volatile markets provides drawdown protection to such strategies.

The modified portfolio has an information ratio (i.e., excess return adjusted for its variance) of 2.9 when the base strategy is used as the benchmark; the value added is very consistent over time, as shown in Exhibit 5.

Our approach is similar to that employed by Israelov and Katz [2011] in a long-only country allocation context; that study also used reversals to time value and momentum strategies and found that the tactically

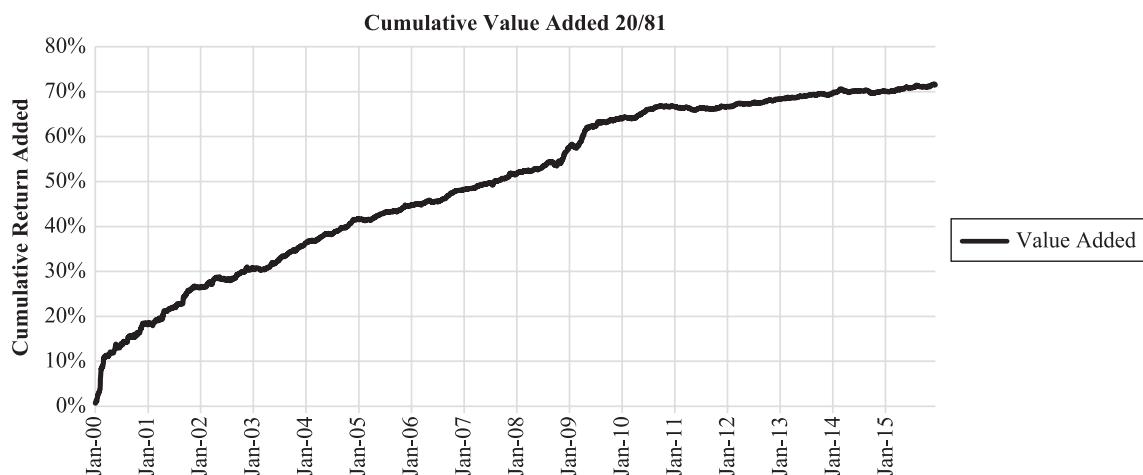
timed portfolio outperforms a blend of short- and long-horizon factors after transaction costs.

Latency

For the more aggressive, "corroborate" type screens, we are reducing our portfolio to those positions where our long- and short-term alphas agree. Because the short-term alphas are realized more quickly, it stands to reason that the value added by these screens depends on our entering into the positions rather quickly, contrary to the long-horizon nature of the portfolio. In Exhibit 6, we look at what happens if we wait an additional 1, 2,

EXHIBIT 5

Cumulative Value Added for the 20/81 Entry and Exit Points for TM1



Source: ExtractAlpha.

EXHIBIT 6

Results for Lag Days before Using TM1

Lag Days	Strategy	Annualized Gross		Annualized Net		Daily Turnover	Stocks per Side
		Return	Sharpe	Return	Sharpe		
1	Base	10.8%	1.08	5.5%	0.55	6.4%	252
	Long 90, Short 11	38.5%	3.71	33.8%	3.26	5.6%	220
2	Base	10.3%	0.99	5.0%	0.48	6.4%	226
	Long 90, Short 11	33.8%	3.29	29.2%	2.84	5.6%	219
3	Base	10.1%	0.97	4.8%	0.46	6.4%	225
	Long 90, Short 11	31.3%	3.05	26.6%	2.60	5.6%	218
4	Base	10.6%	1.01	5.3%	0.51	6.4%	224
	Long 90, Short 11	29.5%	2.88	24.9%	2.43	5.6%	217

Source: ExtractAlpha.

or 3 days before using the TM1 signal, that is, if using a timing signal is still useful even if the positions are entered into gradually.

Although a long latency mutes the efficacy of timing somewhat, the timed strategy is still dramatically stronger than the base strategy.

Risk

Finally, we examine whether the modified portfolios exhibit significantly different risk exposures, using a standard fundamental-style risk model, and find that they do not. Exhibit 7 plots the long and short portfolios' exposures to risk factors on a z -score scale; the profiles are essentially the same for all variants.

Therefore, it appears that TM1 can be layered into an existing fundamental quantitative strategy and provide consistently improved entry and exit tactics without changing the fundamental risk or turnover characteristics of the portfolio, simply by being used as a pre-trade screen.

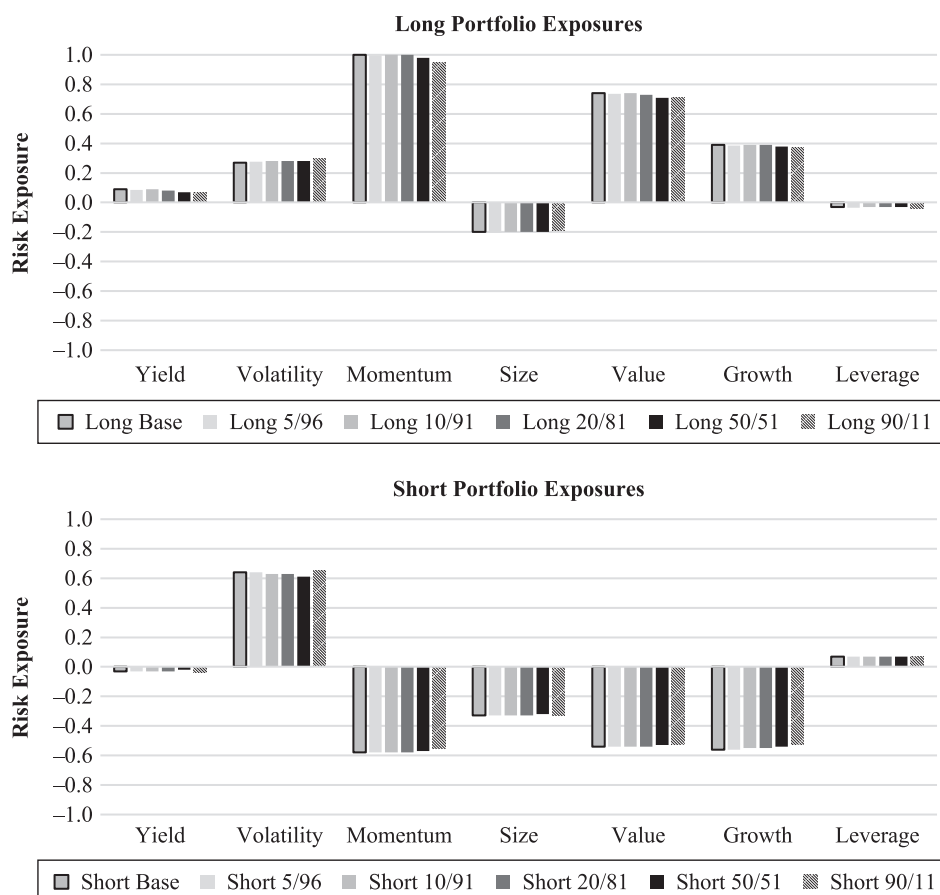
OTHER APPROACHES

These approaches are only examples to demonstrate possible uses of short-horizon signals as a tactical position management tool. A particular manager's implementation may suggest different use cases, such as the following:

- *Going down the list.* Rather than delaying a trade that is contradicted by the short-horizon alpha,

EXHIBIT 7

Exposure to Risk Factors for Long and Short Portfolios, Using z -Score Scale



Source: ExtractAlpha.

a manager could simply take a candidate that is slightly weaker than usual in terms of base strategy alpha but that is not contradicted by the short-horizon alpha. This approach could be implemented at frequencies slower than daily, such as a monthly rebalance, because it does not require waiting until the next rebalance. In practice, in our decile implementation, this means that the modified candidates are generally within the top and bottom 12 percentiles rather than the top and bottom 10, so we are not going very far down the list. We have found that this method provides a similar level of value added to our main “delay” approach.

- *Execution algorithms* could be set to a more aggressive level if the short-horizon alpha and the manager’s desired position are aligned (say, entering over the course of the first day) and to a less aggressive level if they do not (say, entering positions over multiple days).
- The short-horizon alpha could also be used in a *penalty function* in determining position sizes when the base strategy alphas are compared with estimated risks and costs.
- Instead of blending alphas, Garleanu [2013] suggested *blending portfolios*: The portfolio’s asset weights are in-between the optimal current portfolio (incorporating the short-term alphas) and where they will be in the future (which will be more dependent on the long-term alphas).

Short-horizon factors other than TM1 may have value as timing signals; we’ve found in several other studies, however, that factors such as earnings revisions do not produce the same level of value added. It appears that TM1 particularly benefits from being uncorrelated to long-horizon strategies in terms of when it is most effective; its performance time series has a negative correlation to that of most long-horizon factors, unlike earnings revisions, for example, which tend to work when momentum works. Furthermore, technical factors tend to be effective in high-volatility environments, unlike some long-horizon strategies (Jha [2016b]). It is this time series diversification that seems to provide much of the benefit of our timing methodology.

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